

PHASE 1 – BRAINSTORMING AND IDEATION

1. PROJECT TITLE

Script-Controlled ACL – Restrict Record Access Based on Field Value

2. PROJECT OVERVIEW

The proposed project focuses on implementing a **Script-Controlled Access Control List (ACL)** in the ServiceNow platform to provide controlled access to records based on predefined conditions.

The system is designed around an **Institution Details** table containing student information such as student name, faculty name, branch, email, phone number, and description.

The primary access-control requirement is to restrict record access based on the **Branch** field. In the proposed scenario, EEE-related records are accessible to authorized EEE users, while administrators retain full access.

In addition to record-level restrictions, separate ACLs are used to control the **Read, Create, Write, and Delete** operations through custom roles.

3. PROBLEM IDENTIFICATION

Organizations and educational institutions often maintain records that should not be accessible to every user.

A simple role-based permission model may not be sufficient when access must also depend on the values stored within an individual record. For example, users belonging to one academic branch may need access only to records associated with their branch.

Without appropriate record-level security, unauthorized users may be able to:

- View restricted records.
- Create records without the required permission.
- Modify existing records.
- Delete records.

Therefore, a mechanism is required to evaluate both **user authorization** and **record-specific conditions** before allowing access.

4. PROPOSED SOLUTION

The proposed solution uses **ServiceNow Script-Controlled ACLs** to implement record-level security.

A custom table named **Institution Details** is created to store student information. The table contains a **Branch** field with the values:

- ECE
- EEE
- CSE

Custom roles are created to control different operations:

Custom Role	Access Operation
bb1	Read
bb2	Create
bb3	Write
bb4	Delete

The ACL configuration evaluates the applicable user permissions and record conditions before granting access.

Administrators are provided with full access according to the project requirement.

5. PROJECT OBJECTIVES

The primary objectives of the project are:

1. To implement a Script-Controlled ACL in ServiceNow.
2. To restrict access to records based on the **Branch** field.
3. To implement role-based permissions using custom roles.
4. To control **Read, Create, Write, and Delete** operations independently.
5. To prevent unauthorized users from accessing protected records.
6. To provide administrators with full access to the configured records.
7. To demonstrate the practical implementation of ServiceNow record-level security.

6. PROPOSED SYSTEM

The proposed system consists of the following major components:

6.1 User Management

A test user representing an EEE user is created in ServiceNow. The required custom roles are assigned to the user based on the access level being tested.

6.2 Custom Roles

Four custom roles are created:

- bb1 – Read permission
- bb2 – Create permission
- bb3 – Write permission
- bb4 – Delete permission

6.3 Institution Details Table

A custom table named `u_institution_details` is created to store student and institution-related information.

6.4 Script-Controlled ACLs

Separate ACLs are configured for:

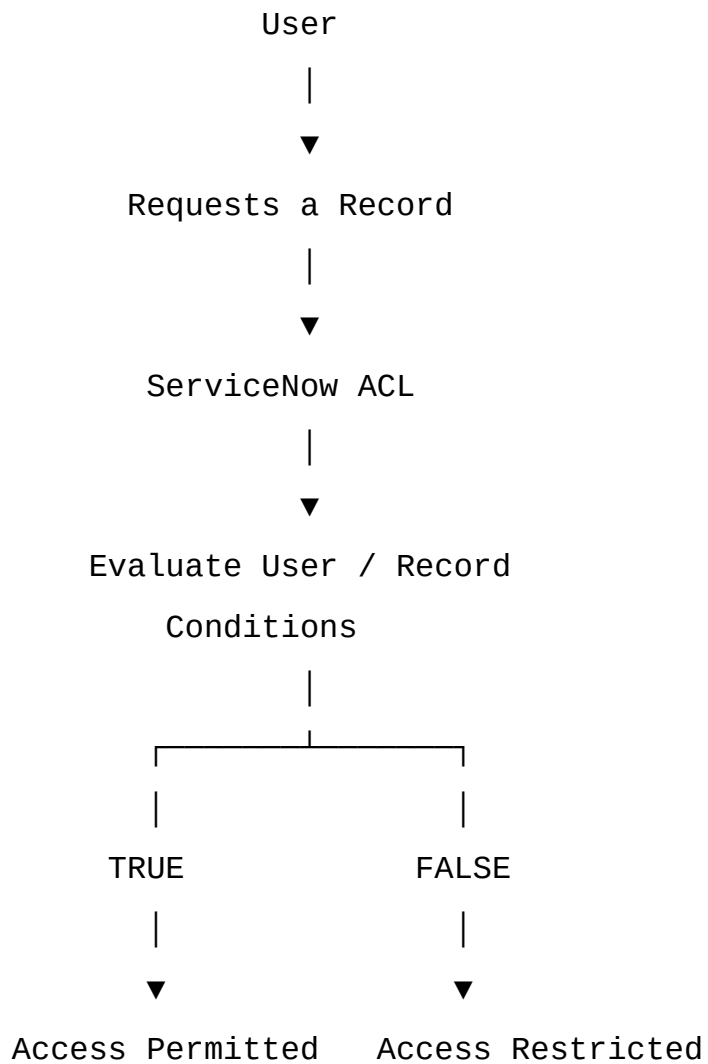
- Read
- Create
- Write
- Delete

6.5 Access Verification

The system is tested using different users and different branch values to verify whether the configured ACLs enforce the intended access restrictions.

7. ACCESS-CONTROL MODEL

The overall access-control process is represented below:



The ACL determines whether the requested operation is permitted based on the configured security conditions.

8. ACCESS PERMISSION MODEL

The project demonstrates progressive access through multiple custom roles.

User Role	Read	Create	Write	Delete
bb1	✓	—	—	—
bb1 + bb2	✓	✓	—	—
bb1 + bb2 + bb3	✓	✓	✓	—
bb1 + bb2 + bb3 + bb4	✓	✓	✓	✓
admin	✓	✓	✓	✓

9. PROPOSED DATA STRUCTURE

The project uses the following custom table:

Table Label: Institution Details

Table Name: u_institution_details

Field	Type	Purpose
Student Roll Number	Auto Number	Unique student record identification
Student Name	Reference – User	Identifies the student
Faculty Name	Reference – User	Identifies the faculty member
Branch	Choice	Stores ECE, EEE, or CSE
Email	String	Stores email information
Phone Number	String	Stores contact information
Description	Multi-line String	Stores additional details

The **Branch** field is the primary record attribute used in the access-control scenario.

10. ACCESS-CONTROL SCENARIO

The project uses the following scenario to demonstrate ACL behavior:

Only authorized EEE users should be able to access the applicable EEE records, while administrators retain full access.

Records are created with different branch values such as ECE, EEE, and CSE.

The ACL configuration is then tested using users with different roles.

Expected behavior

User	Expected Access
EEE user with required role	Access to permitted EEE records
User without required role	Access restricted
Administrator	Full access

The exact effective permissions are determined by the ACL configuration implemented in the ServiceNow instance.

11. PROJECT SCOPE

The scope of the project includes:

- Creating the required ServiceNow users.
- Creating custom roles.
- Assigning roles to test users.
- Creating the Institution Details table.
- Creating required fields.
- Creating records with different branch values.
- Configuring Read ACL.
- Configuring Create ACL.
- Configuring Write ACL.
- Configuring Delete ACL.
- Testing the ACLs using different users.
- Verifying administrator access.
- Documenting the implementation and results.

The project is limited to demonstrating record-level access control within the selected ServiceNow environment.

12. EXPECTED OUTCOME

The expected outcome is a functional ServiceNow security configuration in which access to the Institution Details records is controlled through ACLs.

The completed implementation should demonstrate that:

- Record access can be controlled using a scripted ACL.
- Branch information can be incorporated into access-control decisions.
- Different operations can be controlled independently.
- Custom roles can be used to define user permissions.
- Unauthorized users are restricted from protected operations.
- Administrators retain full access.

13. BENEFITS

The proposed solution provides the following benefits:

- **Improved Security:** Prevents unauthorized access to protected records.
- **Record-Level Control:** Allows access decisions to consider individual record values.
- **Role-Based Permissions:** Provides separate permissions for different operations.
- **Flexibility:** Allows custom access-control logic through scripting.
- **Controlled Operations:** Read, Create, Write, and Delete operations can be managed independently.
- **Practical Implementation:** Demonstrates a real-world ServiceNow security use case.

14. FUTURE ENHANCEMENTS

The project can be extended with additional security capabilities, including:

- Department-based access restrictions.
- Multiple field-based access conditions.
- Dynamic group-based authorization.
- Additional user attributes in ACL evaluation.
- Security audit and access logging.
- More granular role and permission management.
- Integration with additional ServiceNow security features.

15. CONCLUSION

The Brainstorming and Ideation Phase establishes the foundation for the **Script-Controlled ACL – Restrict Record Access Based on Field Value** project.

The proposed solution uses ServiceNow ACLs, custom roles, and record-level conditions to control access to Institution Details records.

The project specifically demonstrates how **Read, Create, Write, and Delete** permissions can be controlled independently while applying a branch-based access restriction and maintaining full administrative access.

The identified objectives, scope, and access-control requirements provide the foundation for the subsequent **Requirement Analysis Phase**.